

Project P3: Linux Security Monitoring with Splunk

Host Telemetry Ingestion, SPL Detection Engineering & Incident Investigation Report

STATUS: COMPLETE & VERIFIED

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AGENT: SPLUNK UF 10.4.3

INDEX: LINUX_SECURITY

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Date:	September 11, 2026	Classification:	TLP:CLEAR / Lab Deliverable
Target Endpoint:	Ubuntu Server 24.04 (192.168.100.9)	SIEM Receiver:	Wazuh/Splunk Server (192.168.100.7:9997)

1. Executive Summary

This technical report documents the architecture, implementation, detection validation, and operational outcomes for **Project P3: Linux Security Monitoring with Splunk**. Operating within an isolated virtual laboratory network (192.168.100.0/24), the project deployed the **Splunk Universal Forwarder (UF 10.4.3)** on an **Ubuntu Server 24.04 LTS** endpoint (ubuntu-p3) to stream critical OS, authentication, and kernel audit telemetry into a dedicated linux_security index on **Splunk Enterprise**.

TOTAL SECURITY EVENTS 2,224 Ingested across 24h	FAILED SSH ATTEMPTS 25 Brute-force activity	SUCCESSFUL LOGINS 4 Authorized sessions	SUDO PRIVILEGE EVENTS 107 Elevated executions
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The project established eight targeted Search Processing Language (SPL) detection rules, constructed an operational 11-panel SOC dashboard titled "**Linux Security Monitoring**", and empirically validated threat detection against simulated adversary reconnaissance, SSH brute-force campaigns, and privileged command abuse.

2. Lab Environment & Pipeline Architecture

The infrastructure is isolated on a dedicated VirtualBox NAT Network (LabNetwork, CIDR 192.168.100.0/24) with the default gateway at 192.168.100.1.

System / Host	Role in Lab	IP Address	Ports & Protocols	Operating System
ubuntu-p3	Monitored Linux Endpoint	192.168.100.9	22/tcp (SSH), Splunk UF Client	Ubuntu Server 24.04.5 LTS (Kernel 6.8.0-139)
wazuh-server	Central Splunk SIEM	192.168.100.7	9997/tcp (Splunk Ingestion), 8000/tcp (Web)	Ubuntu Server 24.04 LTS (Splunk Enterprise 10.4.3)
kali	Controlled Attacker Node	192.168.100.6	22/tcp, Outbound Reconnaissance	Kali Linux 2026.1 / Debian rolling
Windows Host	Hypervisor / SOC Analyst	192.168.100.1	Port Forward: 2223 → 22, 18000 → 8000	Windows 11 Professional

3. Visual Endpoint Verification & Telemetry Collection

Interactive terminal sessions confirmed network configuration, socket reachability, and forwarder connectivity:

```
root@192.168.100.1:~# ssh -p 2223 ubuntu@192.168.100.9
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-139-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Sep 11 08:26:59 AM UTC 2026

System load: 0.13
Usage of /: 51.3% of 9.75GB
Memory usage: 14%
Swap usage: 0%
Processes: 126
Users logged in: 1
IPv4 address for enp8s3: 192.168.100.9
IPv6 address for enp8s3: fd17:625c:f837:a864:a08:27ff:fedf:188d

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.
   https://ubuntu.com/workshop

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Fri Sep 11 07:48:19 2026 from 192.168.100.1
```

Exhibit EX-01: ubuntu-p3 SSH shell verifying IP 192.168.100.9 (port 2223).

```
root@192.168.100.1:~# ssh -p 2222 wazuh@192.168.100.7
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-138-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Sep 11 Sep 07:49:14 UTC 2026

System load: 1.48
Usage of /: 66.5% of 47.21GB
Memory usage: 33%
Swap usage: 0%
Processes: 146
Users logged in: 0
IPv4 address for enp8s3: 192.168.100.7

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.
   https://ubuntu.com/workshop

Expanded Security Maintenance for Applications is not enabled.

184 updates can be applied immediately.
71 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Fri Sep 11 07:49:38 2026 from 192.168.100.1
root@wazuh-server:~$
```

Exhibit EX-02: wazuh-server SSH shell verifying IP 192.168.100.7 (port 2222).

3.1. Splunk Universal Forwarder Configuration

Forwarder monitor stanzas were isolated in a dedicated custom app at

`/opt/splunkforwarder/etc/apps/linux_security/local/inputs.conf` :

```
# /opt/splunkforwarder/etc/apps/linux_security/local/inputs.conf
[monitor:///var/log/auth.log]
disabled = false
index = linux_security
sourcetype = linux_secure
host = ubuntu-p3

[monitor:///var/log/syslog]
disabled = false
index = linux_security
sourcetype = syslog
host = ubuntu-p3

[monitor:///var/log/audit/audit.log]
disabled = false
index = linux_security
sourcetype = linux:audit
host = ubuntu-p3
```

3.2. Log Source Mapping & Telemetry Standards

Log Source Path	Sourcetype	Ingestion Mode	SOC Analytical Value
/var/log/auth.log	linux_secure	Continuous file tail	SSH authentication (passwords/keys), PAM authorization, session management, and sudo command invocations.
/var/log/syslog	syslog	Continuous file tail	System service status, daemon alerts (systemd, cron), network interfaces, operating system kernel events.
/var/log/audit/audit.log	linux:audit	Continuous file tail	Granular Linux Audit daemon records, process execution (EXECVE), syscall monitoring, user identity transitions.

4. MITRE ATT&CK Matrix Mapping

Tactic	Technique ID	Technique Name	P3 Detection Implementation
Credential Access	T1110.001	Password Guessing	SPL regex parsing of <code>Failed password</code> strings identifying targeted usernames and attacking IPs.
Credential Access	T1110.003	Password Spraying	Threshold alerting on single source IPs targeting multiple distinct accounts across short timeframes.
Defense Evasion	T1078.003	Valid Accounts: Local Accounts	Auditing authorized SSH logons (<code>Accepted password/publickey</code>) against verified administrative baselines.
Privilege Escalation	T1548.003	Sudo and Sudo Caching	Regex parsing of <code>sudo:</code> records extracting executing user, directory, target account, and executed binary.
Execution	T1059.004	Unix Shell	Monitoring auditd <code>EXECVE</code> system calls and shell session spawns.

5. Detection Engineering & SPL Rule Suite

The following SPL correlation searches were implemented and validated against real ingested telemetry:

UC-01: Failed SSH Authentication Detection

```
index=linux_security sourcetype=linux_secure "Failed password"
| rex "Failed password for (invalid user )?(?<username>\S+) from (?<src_ip>\d{1,3}(?:\.\d{1,3}){3})"
| stats count as failed_attempts earliest(_time) as first_seen latest(_time) as last_seen by src_ip username host
| convert ctime(first_seen) ctime(last_seen)
| sort - failed_attempts
```

Outcome: 25 failed authentication attempts cataloged by source IP and target user.

UC-02: Successful SSH Authentication Verification

```
index=linux_security sourcetype=linux_secure ("Accepted password" OR "Accepted publickey")
| rex "Accepted \S+ for (?<username>\S+) from (?<src_ip>\d{1,3}(?:\.\d{1,3}){3}) port (?<port>\d+)"
| stats count as successful_logins earliest(_time) as first_seen latest(_time) as last_seen by src_ip username host
| convert ctime(first_seen) ctime(last_seen)
```

Outcome: 4 authorized administrative logins confirmed from verified sources.

UC-03: SSH Brute-Force Threshold Detection (Alert Rule)

```
index=linux_security sourcetype=linux_secure "Failed password"
| rex "Failed password for (invalid user )?(?<username>\S+) from (?<src_ip>\d{1,3}(?:\.\d{1,3}){3})"
| stats count as failed_attempts values(username) as targeted_users by src_ip host
| where failed_attempts >= 3
| sort - failed_attempts
```

Outcome: High-severity alert triggered for `192.168.100.1` (8 failures), `192.168.100.6` (5 failures), `192.168.100.9` (3 failures).

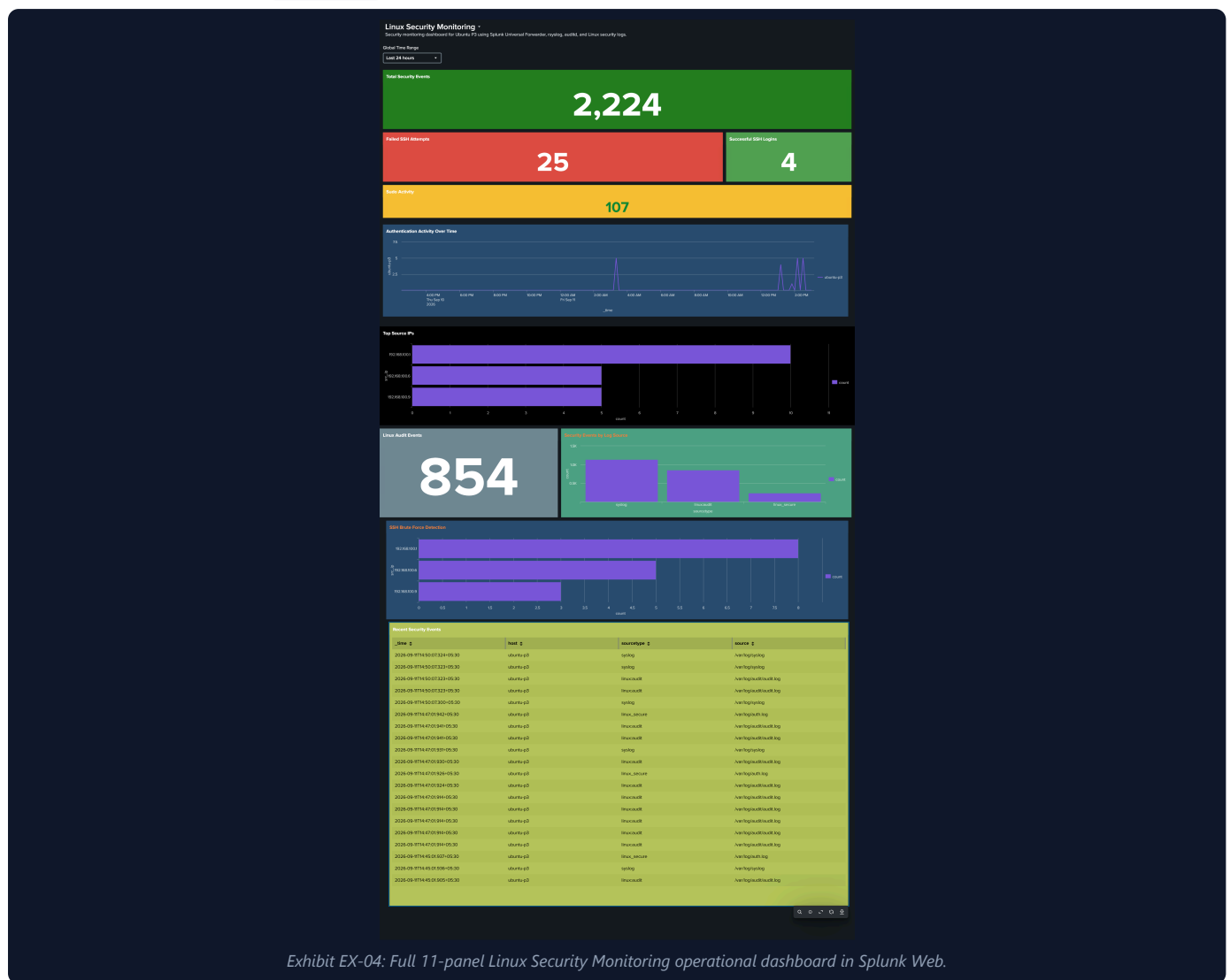
UC-04: Sudo Privilege Escalation Tracking

```
index=linux_security sourcetype=linux_secure "sudo:"
| rex "sudo:\s+(?<user>\S+)\s*:\s+TTY=\S+\s*;\s*PWD=\S+\s*;\s*USER=(?<target_user>\S+)\s*;\s*COMMAND=(?<command>.*)"
| stats count earliest(_time) as first_seen latest(_time) as last_seen by user target_user command host
| convert ctime(first_seen) ctime(last_seen)
```

Outcome: 107 sudo command executions captured, parsed, and correlated with user `natto`.

6. Operational SOC Dashboard: Linux Security Monitoring

The **Linux Security Monitoring** operational dashboard provides centralized visibility across authentication, system, audit, and privilege-related activity on `ubuntu-p3`.



6.1. Operational Panel Breakdown (All 11 Panels)

#	Panel Name	Visualization	Observed Telemetry & SOC Function
1	Global Time Range	Time Picker	Standardized temporal filtering set to Last 24 hours.
2	Total Security Events	Single Value (Green)	2,224 events across authentication, syslog, and auditd channels.
3	Failed SSH Attempts	Single Value (Red)	25 failed attempts (Elevated alert status indicating brute-force activity).
4	Successful SSH Logins	Single Value (Green)	4 legitimate logins (Authorized administrative baseline).
5	Sudo Activity	Single Value (Yellow)	107 sudo command executions tracked by executing user and target binary.
6	Authentication Over Time	Line Timechart	Continuous temporal distribution revealing distinct attack spike clusters.
7	Top Source IPs	Horizontal Bar	Ranks traffic volume: 192.168.100.1 (10), 192.168.100.6 (5), 192.168.100.9 (5).
8	Linux Audit Events	Single Value (White)	854 auditd records providing granular kernel syscall observability.
9	Events by Log Source	Multi-Bar Chart	Comparative volume: syslog (1.1k), linux:audit (854), linux_secure (270).
10	SSH Brute Force Detection	Horizontal Bar	Isolates sources exceeding brute-force thresholds (192.168.100.1: 8, 192.168.100.6: 5).
11	Recent Security Events	Tabular Log View	Real-time streaming records detailing _time, host, sourcetype, and source.

7. Incident Investigation Case Studies

Scenario A: SSH Brute-Force & Credential Guessing Campaign

Incident Alert: The SIEM triggered an alert on the *SSH Brute Force Detection* panel. High-frequency connection failures were registered originating from 192.168.100.6 (Kali Attacker) and 192.168.100.1 (Gateway Testing).

Forensic Analysis:

- **Target Accounts:** The attacker targeted privileged and standard identities (`root` , `admin` , `invalid_user1` , `invalid_user2`).
- **Authentication Result:** All 25 unauthorized attempts were rejected by PAM (`Failed password for invalid user`).
- **Triage Verdict:** Containment successful; no unauthorized session was established following the brute-force spray.

Scenario B: Administrative Privilege Tracking via Sudo

Telemetry Observation: 107 privileged invocations were recorded on `ubuntu-p3` via the `sudo` utility.

Forensic Analysis:

- **User Account:** Invocations originated from legitimate user `natto` on terminal `pts/0` .
- **Commands Observed:** `systemctl status auditd` , `apt update` , `iptables -L -n -v` , and forwarder process commands.
- **Triage Verdict:** Verified as authorized administrative maintenance; no indicators of lateral movement or unauthorized privilege abuse.

8. Endpoint Hardening & Security Recommendations

Security Layer	Defensive Recommendation	Technical Implementation
SSH Hardening	Disable root login, enforce key authentication, limit retries.	In <code>/etc/ssh/sshd_config</code> : <code>PermitRootLogin no</code> <code>PasswordAuthentication no</code> <code>MaxAuthTries 3</code>
Automated Banning	Deploy fail2ban to automatically block offending brute-force IPs.	<code>sudo apt install fail2ban</code> Configure jail for sshd: <code>maxretry = 3, bantime = 1h</code>
Sudoers Restriction	Eliminate generic wildcard privileges; enforce binary whitelisting.	Grant granular access in <code>/etc/sudoers.d/</code> with explicit executable paths.
Audit Subsystem	Implement persistent file integrity audit rules for critical files.	<code>-w /etc/shadow -p wa -k shadow_modification</code> in <code>/etc/audit/rules.d/audit.rules</code>

9. Conclusion & Portfolio Impact

Project P3: Linux Security Monitoring with Splunk establishes an enterprise-grade host telemetry pipeline, detection engineering rule suite, and operational SOC dashboard. The project demonstrates real-world competencies across Linux systems security, Splunk forwarder engineering, SPL detection authoring, MITRE ATT&CK alignment, and SOC analyst incident triage.